

Network Troubleshooting, Management, Monitoring, and Optimization

1) Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Step 1: Run the Command

- Open Command Prompt.
- Type the following command:
`ipconfig /all`
- Press Enter and locate the active network adapter.

Result

- IPv4 Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server 1: 8.8.8.8
- DNS Server 2: 1.1.1.1

2) Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Step 1: Disconnect the Network

- Disconnect Wi-Fi or unplug the Ethernet cable.
- Open Command Prompt and run:
`ping www.google.com`
- Record the result.

Step 2: Reconnect and Test Again

- Reconnect Wi-Fi or the Ethernet cable.
- Run the same command again:
`ping www.google.com`
- Compare the new result with the disconnected result.

Result

- Before reconnection: Request timed out / Destination host unreachable (sample result)
- After reconnection: Reply received successfully, with response times shown in ms (sample result)

3) Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Step 1: Monitor Network Usage

- Open Task Manager using Ctrl + Shift + Esc, or open Resource Monitor.
- Open the Performance/Network view or the relevant network monitoring section.
- Stream a YouTube video for approximately 5 minutes.

Result

- Peak network usage: Approximately 25 Mbps (sample result)
- Process using the most bandwidth: Chrome / browser process (sample result)
- Observation: Network usage increased while the YouTube video was buffering and streaming (sample result)

4) Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Step 1: Capture Traffic

- Open Wireshark and select your active network interface.
- Start the capture and browse Flipkart or Instagram Web for at least 30 seconds.
- Stop the capture after the required time.

Step 2: Apply the Filter

- Use the Wireshark display filter bar.
- Try a filter such as:
`http`
- For HTTPS/TLS traffic, use:

`tls`

Step 3: Information Visible in Packets

- 1. Source and destination IP addresses.
- 2. Transport-layer information such as TCP source/destination ports and packet details.
- 3. Protocol, packet length, timing, and TLS/HTTP-related metadata visible in the capture.

Result

- Filter used: `tls` (HTTPS/TLS traffic)
- Information 1: Source and destination IP addresses
- Information 2: TCP source/destination ports and packet details
- Information 3: Packet length, timing, and TLS-related metadata

5) Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Troubleshooting Checklist

- 1. Check Internet speed and latency using a reliable speed test.
- 2. Prefer Ethernet over Wi-Fi for gaming when possible.
- 3. If using Wi-Fi, use the 5 GHz/6 GHz band when supported and stay close to the router.
- 4. Restart the router and check whether other devices are consuming high bandwidth.
- 5. Close unnecessary downloads, streaming applications, and background updates.
- 6. Check for router firmware updates and keep the computer/network drivers updated.
- 7. If available, enable suitable QoS/game-prioritization features on the router.